

AI-Based Helmet Violation Detection and Automatic E-Challan Generation Using YOLOv11, SAM 2 and OCR

Course: Machine Learning Section 4

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Abstract

This project proposes an AI-based helmet violation detection and e-challan management system using YOLOv11, OCR, and a distributed backend. The system detects helmet violations and vehicle number plates from traffic camera footage, extracts the plate number using PaddleOCR, and stores vehicle, violation, and challan details in a MySQL database. A REST API connects the AI module with the backend, enabling automatic violation recording, challan generation, and efficient retrieval of records. The project demonstrates database design, SQL, transaction management, API communication, and distributed backend development.

□ Workflow (high level): Traffic Video → YOLOv11 → SAM 2 → PaddleOCR → FastAPI → MySQL → E-Challan



Problem Statement



Current Challenges

- Manual monitoring of traffic violations demands continuous human effort
- Helmet violations are hard to track at scale
- Manual number-plate identification is slow and error-prone
- Record maintenance lacks structure — retrieval is inefficient
- Challan processing is often separate and manual

Project Goal

Automatically detect helmet violations, extract vehicle numbers, persist structured records in **MySQL**, and manage e-Challan lifecycle via a FastAPI backend and admin dashboard for rapid retrieval and auditing.

Manual Monitoring → Time consuming → Human effort → Difficult records

Proposed → Automated detection → API → **MySQL** → E-Challan

Existing System vs Proposed System

Existing System

Manual traffic monitoring

Human-based helmet identification

Manual number plate checking

Manual record maintenance

Slow retrieval and separate challan processing

Proposed System



YOLOv11 – Detection

Real-time motorcycle, helmet and plate detection.



SAM 2 – Segmentation

Accurate region masks for better OCR and evidence capture.



PaddleOCR – Recognition

Robust plate-to-text conversion for multiple formats.



FastAPI → MySQL

Backend API accepts detections and persists structured records.

Existing: Human → Manual Records | Proposed: AI → API → **MySQL** → E-Challan

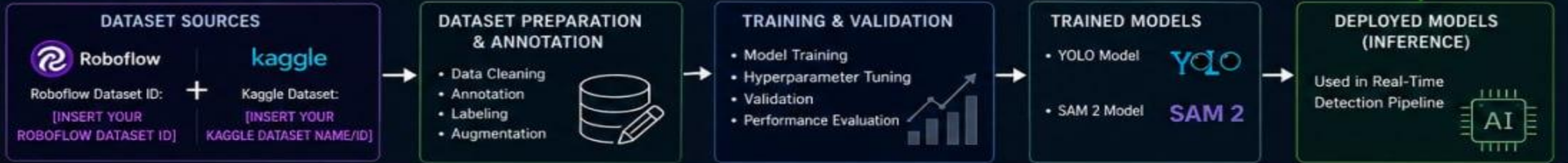
SYSTEM ARCHITECTURE

Helmet Violation Detection and E-Challan Generation System

REAL-TIME DETECTION PIPELINE



TRAINING / DATASET PIPELINE



→ Data Flow

-----> Training Flow

↔ Database Interaction

Tools Used to Implement the System

AI / Computer Vision

- **Python** — primary implementation language
- **YOLOv11** — detection (motorcycle, helmet, number plate)
- **SAM 2** — precise segmentation masks
- **PaddleOCR** — license plate text extraction
- **OpenCV** — preprocessing & frame handling

Backend & Database

- **FastAPI** — REST API, validation, and distributed backend endpoints
- **MySQL** — central data-management layer: tables, relationships, SQL queries, transactions
- **React.js** — admin dashboard for verification, challan management and reporting

Hardware used

- Laptop/Desktop with a 8GB VRAM GPU
- NVIDIA RTX 4060 GPU(Our Component)
- CCTV/Traffic Camera

Development & tooling: Roboflow (annotation/dataset), Git/GitHub (VCS), VS Code (IDE).

❏ Final statement: **AI** detects the violation, **OCR** identifies the vehicle, **FastAPI** connects components, and **MySQL** manages complete violation & challan data.

Dataset & Model



Motorcycle Detection

Images annotated for motorcycles across varied angles, lighting and occlusion.



Helmet / No-Helmet

Binary labels per rider; samples include partial occlusion and different helmets.



License Plate

Crops and bounding boxes for plate localisation; prepared for OCR recognition.



Triple-Riding

Annotated examples to enable future multi-rider detection and policy enforcement.

Datasets merged and converted into a unified YOLO format. YOLOv11 is trained to detect motorcycles, riders, helmets and plates; PaddleOCR handles character recognition from cropped plates.

Conclusion

The Helmet Violation Detection and E-Challan Generation System provides an automated approach to identifying helmet violations using Artificial Intelligence and Computer Vision. The system uses YOLO for object detection and SAM 2 for precise image segmentation and masking. OpenCV is used for image and video processing, while OCR extracts vehicle registration numbers from detected license plates. The detected violation details, vehicle number, date, time, and evidence can be stored in a MySQL database and used for the e-challan generation process. Datasets from Roboflow and Kaggle support the development and training of the detection models. Overall, the proposed system can reduce manual monitoring, improve detection efficiency, minimize human errors, and support faster traffic-rule enforcement through an intelligent and scalable AI-based solution.





THANK YOU

QUESTIONS & DISCUSSION

An intelligent and scalable approach towards
automated traffic violation monitoring and enforcement.

AI



NO HELMET
VIOLATION DETECTED



CAPTURE



DETECTION
(YOLO)



SEGMENTATION
(SAM 2)



LICENSE PLATE
DETECTION



OCR



DATABASE
(MySQL)



E-CHALLAN
GENERATION